

强化学习基本原理

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强化学习的目标函数

强化学习的目标函数：

$$J(\pi) = E[\sum_{k=t}^{\infty} \gamma^{k-t} r(s_k, a_k)]$$

如果表示从状态 s 经时间步 t 转移到状态 s' 的概率为： $p(s \rightarrow s', t, \pi)$

我们也可以表示折扣状态分布为： $\rho^{\pi}(s') := \int_S \sum_{t=1}^{\infty} \gamma^{t-1} p_1(s) p(s \rightarrow s', t, \pi) ds$

强化学习目标可以写为： $J(\pi_{\theta}) = \int_S \rho^{\pi}(s) \int_A \pi_{\theta}(s, a) r(s, a) da ds$



随机策略梯度理论

强化学习目标可以写为： $J(\pi_\theta) = \int_S \rho^\pi(s) \int_A \pi_\theta(s, a) r(s, a) da ds$

策略梯度理论：

$$\begin{aligned} \nabla_\theta J(\pi_\theta) &= \int_S \rho^\pi(s) \int_A \nabla_\theta \pi_\theta(a | s) Q^\pi(s, a) da ds \\ &= \mathbb{E}_{s \sim \rho^\pi, a \sim \pi_\theta} \left[\nabla_\theta \log \pi_\theta(a | s) Q^\pi(s, a) \right] \end{aligned}$$

策略梯度很简单，尽管分布依赖于参数，但是策略梯度并不依赖于状态的分布

还没有解决的问题：如何估计行为值函数？



随机策略梯度理论

如何估计行为值函数？

$$\begin{aligned}\nabla_{\theta} J(\pi_{\theta}) &= \int_S \rho^{\pi}(s) \int_A \nabla_{\theta} \pi_{\theta}(a | s) Q^{\pi}(s, a) da ds \\ &= \mathbb{E}_{s \sim \rho^{\pi}, a \sim \pi_{\theta}} \left[\nabla_{\theta} \log \pi_{\theta}(a | s) Q^{\pi}(s, a) \right]\end{aligned}$$

1. 如果不用估计值，REINFORCE算法

2. 如果用函数逼近的方法来估计值函数，则称为Actor-Critic框架。

Actor调整策略网络的参数 π_{θ}

Critic调整行为值函数网络的参数 $Q^w(s, a)$

一般情况下，带入一个函数逼近器来代替行为值函数会引入偏差。与策略相容的无偏差的函数逼近器称为相容函数。



随机策略梯度的相容函数

策略梯度：

$$\begin{aligned}\nabla_{\theta} J(\pi_{\theta}) &= \int_S \rho^{\pi}(s) \int_A \nabla_{\theta} \pi_{\theta}(a | s) Q^{\pi}(s, a) da ds \\ &= \mathbb{E}_{s \sim \rho^{\pi}, a \sim \pi_{\theta}} \left[\nabla_{\theta} \log \pi_{\theta}(a | s) Q^{\pi}(s, a) \right]\end{aligned}$$

将真正的行为值函数 $Q^{\pi}(s, a)$ 用函数来逼近，用来逼近的函数应该满足以下相容性条件：

$$Q^w(s, a) = \nabla_{\theta} \log \pi_{\theta}(a | s)^T w$$

W使得如下目标函数最小：
$$\varepsilon^2(w) = E_{s \sim \rho^{\pi}, a \sim \pi_{\theta}} \left[(Q^w(s, a) - Q^{\pi}(s, a))^2 \right]$$



随机 Off-policy Actor-Critic



采样策略为 β 要评估的策略为 π

目标函数为：

$$\begin{aligned} J_{\beta}(\pi_{\theta}) &= \int_S \rho^{\beta}(s) V^{\pi}(s) ds \\ &= \int_S \int_A \rho^{\beta}(s) \pi_{\theta}(a|s) Q^{\pi}(s, a) da ds \end{aligned}$$

Off-policy 策略梯度为：

$$\nabla_{\theta} J_{\beta}(\pi_{\theta}) = E_{s \sim \rho^{\beta}, a \sim \beta} \left[\frac{\pi_{\theta}(a|s)}{\beta_{\theta}(a|s)} \nabla_{\theta} \log \pi_{\theta}(a|s) Q^{\pi}(s, a) \right]$$



随机策略 VS 确定性策略

随机策略：

$$\pi_{\theta}(a|s) = P[a|s; \theta]$$

在状态 s ，动作符合参数为 θ 的概率分布。训练时按照该概率分布采样。

确定性策略：

$$a = \mu_{\theta}(s)$$

在状态 s ，动作是参数和状态的确定性函数

随机策略梯度对状态空间和动作空间积分。

$$\nabla_{\theta} J(\pi_{\theta}) = E_{s \sim \rho^{\pi}, a \sim \pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(a|s) Q^{\pi}(s, a)]$$

确定性策略梯度只对状态空间积分。

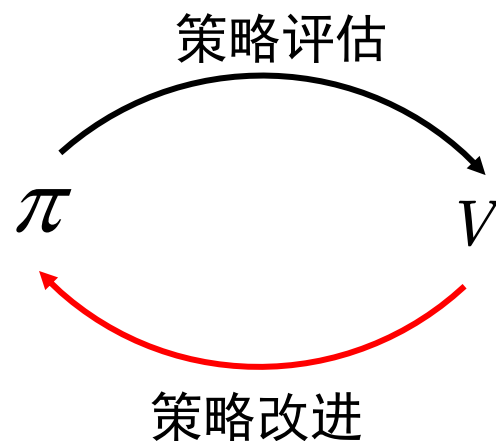
因此，计算随机策略梯度需要更多的样本，特别是当动作空间有很多维数时。



确定性策略梯度强化学习算法



强化学习的框架：广义策略迭代



对于参数化的策略，在进行策略改进时用的是梯度更新



确定性值函数

行为-值函数的定义：

$$Q^{\pi}(s_t, a_t) = \mathbb{E}_{r_{i \geq t}, s_{i > t} \sim E, a_{i > t} \sim \pi} [R_t \mid s_t, a_t]$$

利用迭代方式可以得到：

$$Q^{\pi}(s_t, a_t) = \mathbb{E}_{r_{i \geq t}, s_{t+1} \sim E, [r(s_t, a_t) + \gamma \mathbb{E}_{a_{t+1} \sim \pi} [Q^{\pi}(s_{t+1}, a_{t+1})]]}$$

如果目标策略为确定性策略 μ ，则：

$$Q^{\mu}(s_t, a_t) = \mathbb{E}_{r_t, s_{t+1} \sim E, [r(s_t, a_t) + \gamma Q^{\mu}(s_{t+1}, \mu(s_{t+1}))]}$$

期望仅仅依赖于环境



确定性策略梯度

策略改善：在策略梯度的方向上更新参数

$$\theta^{k+1} = \theta^k + \alpha E_{s \sim \rho^{\mu^k}} [\nabla_{\theta} Q^{\mu^k}(s, \mu_{\theta}(s))]$$

根据链式规则，得到确定性策略梯度理论：

$$\nabla_{\theta} J(\mu_{\theta}) = E_{s \sim \rho^{\mu}} [\nabla_{\theta} \mu_{\theta}(s) \nabla_a Q^{\mu}(s, a)|_{a=\mu_{\theta}(s)}]$$

确定性 Actor-Critic 算法

异策略AC的方法:

行动策略为随机策略 $\beta(s, a)$

要评估的策略为确定性策略: μ_θ

异策略确定性策略梯度: $\nabla_\theta J_\beta(\mu_\theta) = E_{s \sim \rho^\beta} [\nabla_\theta \mu_\theta(s) \nabla_a Q^\mu(s, a)|_{a=\mu_\theta(s)}]$

更新过程:

$$\delta_t = r_t + \gamma Q^w(s_{t+1}, \mu_\theta(s_{t+1})) - Q^w(s_t, a_t)$$

Qlearning(off-policy)

$$w_{t+1} = w_t + \alpha_w \delta_t \nabla_w Q^w(s_t, a_t)$$

$$\theta_{t+1} = \theta_t + \alpha_\theta \nabla_\theta \mu_\theta(s_t) \nabla_a Q^w(s_t, a_t)|_{a=\mu_\theta(s)}$$

确定性策略无需重采样



确定性策略梯度算法中如何评估行为值函数？



定理3:

相容函数逼近



函数逼近器 $Q^w(s, a)$ 是与确定性策略 $\mu_\theta(s)$ 及确定性策略梯度 $E[\nabla_\theta \mu_\theta(s) \nabla_a Q^w(s, a) |_{a=\mu_\theta(s)}]$ 相容条件:

1. $\nabla_a Q^w(s, a) |_{a=\mu_\theta(s)} = \nabla_\theta \mu_\theta(s)^T w$

2. w 最小化如下均方误差:

$$MSE(\theta, w) = E[\varepsilon(s; \theta, w)^T \varepsilon(s; \theta, w)]$$

其中: $\varepsilon(s; \theta, w) = \nabla_a Q^w(s, a) |_{a=\mu_\theta(s)} - \nabla_a Q^\mu(s, a) |_{a=\mu_\theta(s)}$



相容函数的讨论

1. $\nabla_a Q^w(s, a)|_{a=\mu_\theta(s)} = \nabla_\theta \mu_\theta(s)^T w$

满足该条件的相容函数为：

$$Q^w(s, a) = (a - \mu_\theta(s))^T \nabla_\theta \mu_\theta(s)^T w + V^v(s)$$

线性逼近函数对于预测全局的行为值函数并无用处，但是并不妨碍作为局部 critic



相容函数的讨论

2. w 最小化如下均方误差:

$$MSE(\theta, w) = E[\varepsilon(s; \theta, w)^T \varepsilon(s; \theta, w)]$$

其中: $\varepsilon(s; \theta, w) = \nabla_a Q^w(s, a)|_{a=\mu_\theta(s)} - \nabla_a Q^\mu(s, a)|_{a=\mu_\theta(s)}$

直接优化该目标函数不可行: 因为真实的梯度 $\nabla_a Q^\mu(s, a)|_{a=\mu_\theta(s)}$

在实际算法中: 利用线性函数逼近器: $Q^w(s, a) = \phi(s, a)^T w$ 和时间差分的方法。

不能严格满足第二个条件

$$\delta_t = r_t + \gamma Q^w(s_{t+1}, \mu_\theta(s_{t+1})) - Q^w(s_t, a_t)$$

$$w_{t+1} = w_t + \alpha_w \delta_t \nabla_w Q^w(s_t, a_t)$$

$$\theta_{t+1} = \theta_t + \alpha_\theta \nabla_\theta \mu_\theta(s_t) \nabla_a Q^w(s_t, a_t)|_{a=\mu_\theta(s)}$$

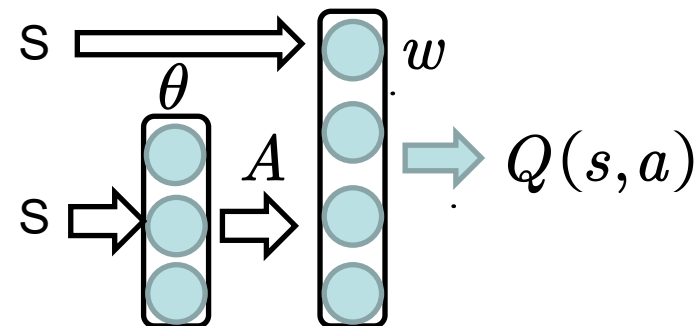
$$Q^w(s, a), \mu_\theta(s)$$

利用神经网络进行逼近时，遇到的问题：

1. 神经网络的训练假设数据之间是独立同分布的，而强化学习顺序采集的数据明显存在相关性。
2. 利用mini批方法进行优化数据可以得到充分利用。

DQN的成功：

1. 经验回放
2. 独立的目标网络



$$\delta_t = r_t + \boxed{\gamma Q^w(s_{t+1}, \mu_\theta(s_{t+1}))} - Q^w(s_t, a_t)$$

$$w_{t+1} = w_t + \alpha_w \delta_t \nabla_w Q^w(s_t, a_t)$$

$$\theta_{t+1} = \theta_t + \alpha_\theta \nabla_\theta \mu_\theta(s_t) \nabla_a Q^w(s_t, a_t)|_{a=\mu_\theta(s)}$$

DDPG

DDPG算法伪代码

Randomly initialize critic network $Q(s, a|\theta^Q)$ and actor $\mu(s|\theta^\mu)$ with weights θ^Q and θ^μ .

Initialize target network Q' and μ' with weights $\theta^{Q'} \leftarrow \theta^Q, \theta^{\mu'} \leftarrow \theta^\mu$

Initialize replay buffer R

for episode = 1, M do

 Initialize a random process $\mathcal{N}$ for action exploration

 Receive initial observation state s_1

 for t = 1, T do

 行动策略为随机策略

 Select action $a_t = \mu(s_t|\theta^\mu) + \mathcal{N}_t$ according to the current policy and exploration noise

 Execute action a_t and observe reward r_t and observe new state s_{t+1}

 Store transition (s_t, a_t, r_t, s_{t+1}) in R

 Sample a random minibatch of N transitions (s_i, a_i, r_i, s_{i+1}) from R

经验回放

 Set $y_i = r_i + \gamma Q'(s_{i+1}, \mu'(s_{i+1}|\theta^{\mu'})|\theta^{Q'})$

目标网络

 Update critic by minimizing the loss: $L = \frac{1}{N} \sum_i (y_i - Q(s_i, a_i|\theta^Q))^2$

 Update the actor policy using the sampled gradient:

$$\nabla_{\theta^\mu} \mu|_{s_t} \approx \frac{1}{N} \sum_i \nabla_a Q(s, a|\theta^Q)|_{s=s_t, a=\mu(s_t)} \nabla_{\theta^\mu} \mu(s|\theta^\mu)|_{s_t}$$

Update the target networks:

$$\begin{aligned} \theta^{Q'} &\leftarrow \tau \theta^Q + (1 - \tau) \theta^{Q'} \\ \theta^{\mu'} &\leftarrow \tau \theta^\mu + (1 - \tau) \theta^{\mu'} \end{aligned}$$

目标网络参数更新

end for
end for

DDPG存在的问题

1. 值函数的过优估计问题

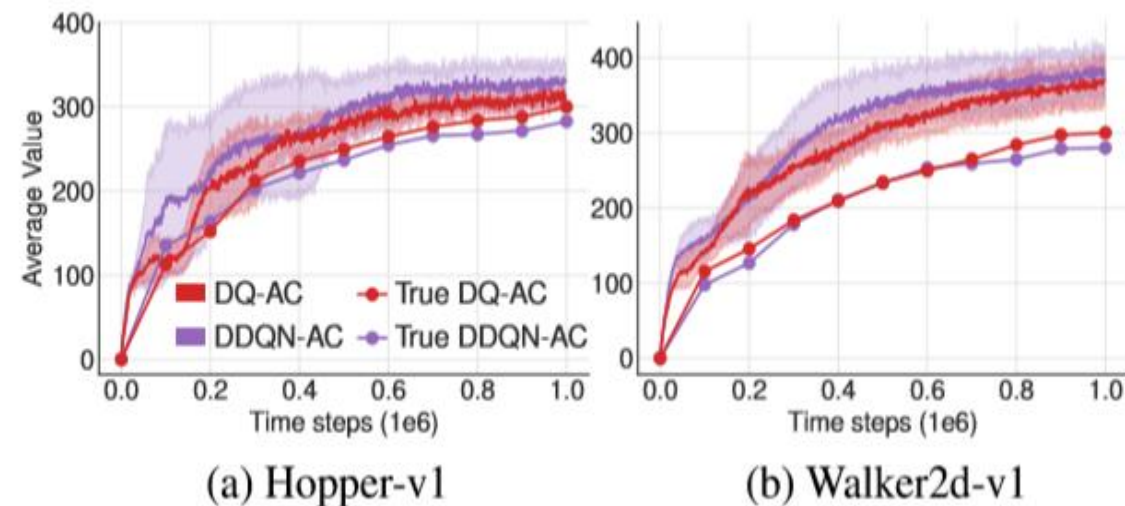
$$\delta_t = r_t + \gamma Q^w(s_{t+1}, \mu_\theta(s_{t+1})) - Q^w(s_t, a_t)$$

$$w_{t+1} = w_t + \alpha_w \delta_t \nabla_w Q^w(s_t, a_t)$$

$$\theta_{t+1} = \theta_t + \alpha_\theta \nabla_\theta \mu_\theta(s_t) \nabla_a Q^w(s_t, a_t)|_{a=\mu_\theta(s)}$$

解决问题的方法：

$$y_1 = r + \gamma \min_{i=1,2} Q_{\theta_i}(s', \pi_{\phi_1}(s'))$$



2. 值函数高的方差

解决方法：独立的目标网络

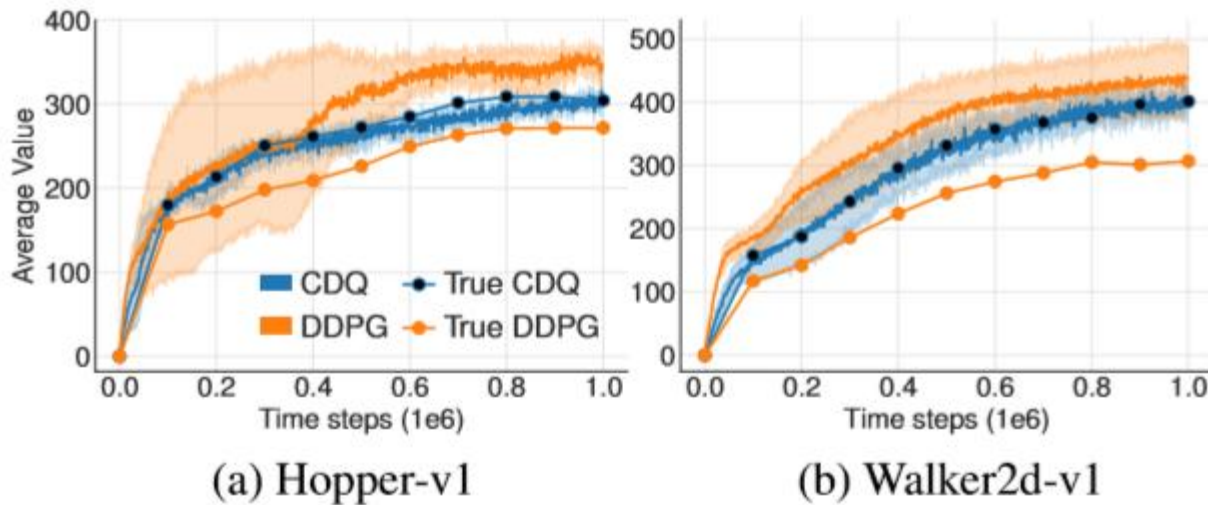
目标策略光滑：

$$y = r + Q_{\theta'}(s', \pi_{\phi'}(s')) + \epsilon,$$

$$\epsilon \sim \text{clip}(\mathcal{N}(0, \sigma), -c, c)$$

3. 策略网络与值函数网络之间存在耦合关系

Twin Delay 双延迟：目标网络更新，策略更新延迟





DDPG扩展TD3

Algorithm 1 TD3

Initialize critic networks $Q_{\theta_1}, Q_{\theta_2}$, and actor network π_ϕ with random parameters θ_1, θ_2, ϕ

Initialize target networks $\theta'_1 \leftarrow \theta_1, \theta'_2 \leftarrow \theta_2, \phi' \leftarrow \phi$

Initialize replay buffer $\mathcal{B}$

for $t = 1$ **to** T **do**

 Select action with exploration noise $a \sim \pi_\phi(s) + \epsilon$,

$\epsilon \sim \mathcal{N}(0, \sigma)$ and observe reward r and new state s'

 Store transition tuple (s, a, r, s') in $\mathcal{B}$

Sample mini-batch of N transitions (s, a, r, s') from $\mathcal{B}$

$\tilde{a} \leftarrow \pi_{\phi'}(s') + \epsilon, \quad \epsilon \sim \text{clip}(\mathcal{N}(0, \tilde{\sigma}), -c, c)$

$y \leftarrow r + \gamma \min_{i=1,2} Q_{\theta'_i}(s', \tilde{a})$

Update critics $\theta_i \leftarrow \text{argmin}_{\theta_i} N^{-1} \sum (y - Q_{\theta_i}(s, a))^2$

if $t \bmod d$ **then**

 Update ϕ by the deterministic policy gradient:

$\nabla_\phi J(\phi) = N^{-1} \sum \nabla_a Q_{\theta_1}(s, a)|_{a=\pi_\phi(s)} \nabla_\phi \pi_\phi(s)$

 Update target networks:

$\theta'_i \leftarrow \tau \theta_i + (1 - \tau) \theta'_i$

$\phi' \leftarrow \tau \phi + (1 - \tau) \phi'$

end if

end for

Environment	TD3	DDPG	Our DDPG	PPO	TRPO	ACKTR	SAC
HalfCheetah	9636.95 $\pm$ 859.065	3305.60	8577.29	1795.43	-15.57	1450.46	2347.19
Hopper	3564.07 $\pm$ 114.74	2020.46	1860.02	2164.70	2471.30	2428.39	2996.66
Walker2d	4682.82 $\pm$ 539.64	1843.85	3098.11	3317.69	2321.47	1216.70	1283.67
Ant	4372.44 $\pm$ 1000.33	1005.30	888.77	1083.20	-75.85	1821.94	655.35
Reacher	-3.60 $\pm$ 0.56	-6.51	-4.01	-6.18	-111.43	-4.26	-4.44
InvPendulum	1000.00 $\pm$ 0.00	1000.00	1000.00	1000.00	985.40	1000.00	1000.00
InvDoublePendulum	9337.47 $\pm$ 14.96	9355.52	8369.95	8977.94	205.85	9081.92	8487.15



D4PG: Distributed Distributional Deep Deterministic Policy Gradient



Actor-critic算法中策略梯度依赖于学到的critic, 这就意味着任何对critic学习过程进行改善的方法将直接改善actor更新质量。

值分布建模了由于函数逼近等带来的随机性, 因此利用值分布进行更新critic将直接改善学习算法的表现。

用到的技巧:

1. 值分布式强化学习更细critic
2. 分布式并行Actor
3. N-step回报
4. 优先经验回访采数据

D4PG: Distributed Distributional Deep Deterministic Policy Gradient

Algorithm 1 D4PG

Input: batch size M , trajectory length N , number of actors K , replay size R , exploration constant ϵ , initial learning rates α_0 and β_0

- 1: Initialize network weights (θ, w) at random
- 2: Initialize target weights $(\theta', w') \leftarrow (\theta, w)$
- 3: Launch K actors and replicate network weights (θ, w) to each actor
- 4: **for** $t = 1, \dots, T$ **do**
- 5: Sample M transitions $(\mathbf{x}_{i:i+N}, \mathbf{a}_{i:i+N-1}, r_{i:i+N-1})$ of length N from replay with priority p_i
- 6: Construct the target distributions $Y_i = \left(\sum_{n=0}^{N-1} \gamma^n r_{i+n} \right) + \gamma^N Z_{w'}(\mathbf{x}_{i+N}, \pi_{\theta'}(\mathbf{x}_{i+N}))$
Note, although not denoted the target Y_i may be projected (e.g. for Categorical value distributions).
- 7: Compute the actor and critic updates

$$\delta_w = \frac{1}{M} \sum_i \nabla_w (Rp_i)^{-1} d(Y_i, Z_w(\mathbf{x}_i, \mathbf{a}_i))$$

$$\delta_\theta = \frac{1}{M} \sum_i \nabla_\theta \pi_\theta(\mathbf{x}_i) \mathbb{E}[\nabla_{\mathbf{a}} Z_w(\mathbf{x}_i, \mathbf{a})] \big|_{\mathbf{a}=\pi_\theta(\mathbf{x}_i)}$$

- 8: Update network parameters $\theta \leftarrow \theta + \alpha_t \delta_\theta, w \leftarrow w + \beta_t \delta_w$
- 9: If $t = 0 \bmod t_{\text{target}}$, update the target networks $(\theta', w') \leftarrow (\theta, w)$
- 10: If $t = 0 \bmod t_{\text{actors}}$, replicate network weights to the actors
- 11: **end for**
- 12: **return** policy parameters θ

D4PG : Distributed Distributional Deep Deterministic Policy Gradient

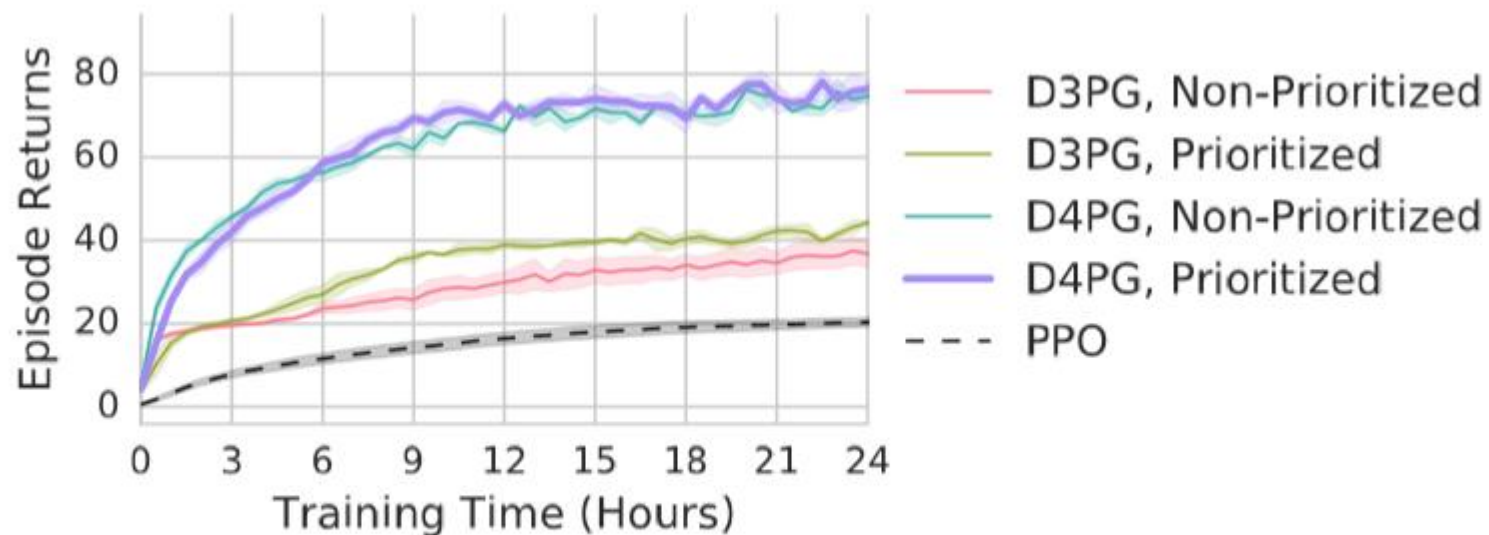


Figure 6: Experimental results for the three-dimensional (humanoid) parkour domain.